

Esercizi

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Risolvere le seguenti equazioni di secondo grado:

1. $x^2 - 9 = 0$
2. $6(x - 1) = x(x + 6)$
3. $(3+x)(x-3) - 2(x+1)^2 = 2 - (x+2)^2 + x^2$
4. $(x - 5)^2 = 25$
5. $(x + 2)^2 = 3$
6. $4(x + 3)^2 = 100$
7. $(2x - 1)(2x + 3) = -3$
8. $\frac{x+1}{2} - \frac{2x-1}{4} = \frac{x^2+1}{4} + \frac{x+2}{4}$
9. $\frac{(x+1)(x-1)}{2} + \frac{x+2}{4} = (x+1)^2 - 1$
10. $x^2 + (x - 2)^2 = (x + 2)^2$
11. $(x + 2)(x - 2) + (x - 1)^2 = x - 3$
12. $2x^2 - 3x - 2 = 0$
13. $4x^2 - 12x + 9 = 0$
14. $x^2 + \frac{5}{2}x - \frac{3}{2}$
15. $x^2 + \frac{6}{5}x + \frac{8}{25} = 0$
16. $(2x-1)(2x+1) + (3x-1)(2x-3) = 6x-1$
17. $-\frac{3}{2} \left(\frac{2}{3}x - \frac{5}{6} \right) = 2x^2 + \frac{7}{2}$
18. $\frac{x-1}{2} + \frac{(x+1)(x-2)}{3} = -\frac{1}{6}(2x-1)^2$

Risolvere le seguenti equazioni di secondo frazionarie:

1. $\frac{1}{x^2 - x} + \frac{1}{x} = 1$
2. $\frac{x^2}{x^2 - 1} + \frac{1}{x + 1} = \frac{1}{x^2 - 1}$
3. $\frac{4}{x - 2} + \frac{x}{x^2 - x - 2} = -\frac{3}{2}$
4. $\frac{(x+1)^2 - 8}{x^2 + 4x + 3} + \frac{1-x}{x+1} = 1 - \frac{3}{x+1}$
5. $\frac{\frac{2(x+2)}{x-2} + \frac{6}{x^2 - 4x + 4}}{\frac{5x - x^2}{x^2 - 4x + 4}} = \frac{x}{x-2} +$
6. $\frac{1+x}{x^2 + 5x + 6} + \frac{1}{x+3} = \frac{1}{2}$
7. $\frac{1}{y^3 - 4y} - \frac{1}{y^2 + 4y + 4} = \frac{2}{3y^2 + 6y}$